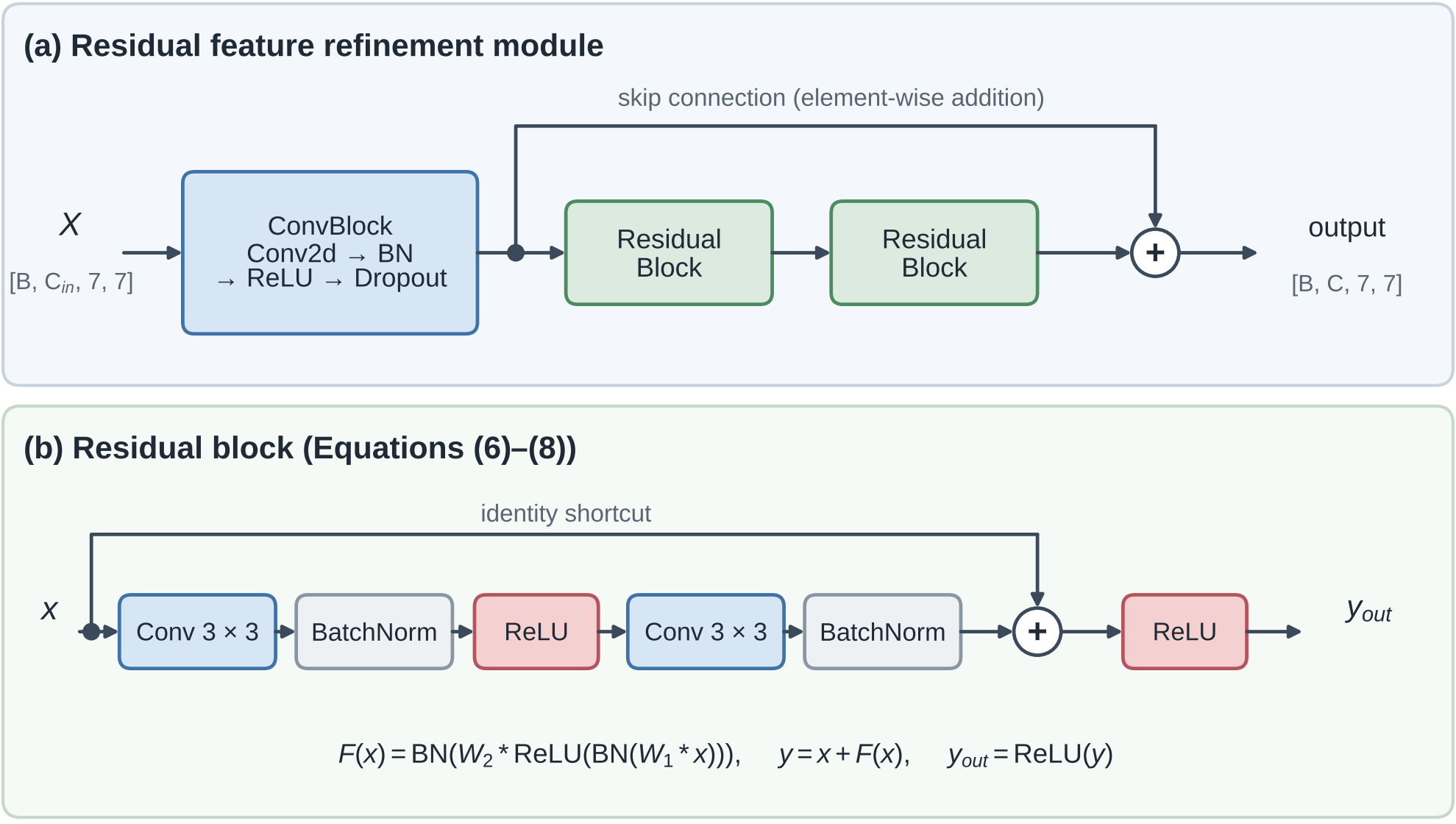
(a) Residual feature refinement module
skip connection (element-wise addition)
X
[B, C_in, 7, 7]
ConvBlock
Conv2d → BN
→ ReLU → Dropout
Residual Block
Residual Block
+
output
[B, C, 7, 7]
(b) Residual block (Equations (6)–(8))
identity shortcut
x
Conv 3 × 3
BatchNorm
ReLU
Conv 3 × 3
BatchNorm
+
ReLU
y_out
F(x) = BN(W_2 * ReLU(BN(W_1 * x))), y = x + F(x), y_out = ReLU(y)